BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE PILANI
Hyderabad Campus
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Mathematical Foundations for Data Science

Tutorial Sheet - 1

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1. Find the Row-Echelon form and Reduced Row-Echelon form of the following matrices using elementary row operations.

   (i) $\begin{bmatrix} 0 & 2 & 2 \\ 1 & 3 & 3 \\ 2 & 4 & 2 \end{bmatrix}$

   (ii) $\begin{bmatrix} 1 & 3 & 3 & 8 & 5 \\ 0 & 1 & 3 & 10 & 8 \\ 0 & 0 & 0 & -1 & -4 \\ 0 & 0 & 0 & 2 & 8 \end{bmatrix}$

   (iii) $\begin{bmatrix} 0 & 1 & 3 & -1 & 2 & 1 \\ 0 & -2 & -6 & 2 & 0 & 2 \\ 0 & 0 & 0 & 1 & -1 & 7 \\ 0 & 2 & 6 & -2 & 2 & 0 \\ 0 & 3 & 9 & -2 & 2 & 7 \end{bmatrix}$

2. Find the inverse of the following matrices by Gauss Elimination.

   (i) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$

   (ii) $\begin{bmatrix} 2 & 1 & -1 \\ 3 & 1 & 0 \\ -1 & 0 & 2 \end{bmatrix}$

   (iii) $\begin{bmatrix} 0 & 1 & -1 & 1 \\ 2 & 2 & 0 & -2 \\ 1 & 1 & -2 & 0 \\ 0 & 1 & 2 & 0 \end{bmatrix}$

3. Let $S = \left\{ \begin{bmatrix} c \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -3 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ c \\ -1 \end{bmatrix} \right\}$.

(i) Find the value/s of c such that S is a linearly dependent set.

(ii) Express the vector $\begin{bmatrix} a_1 \\ a_2 \\ 0 \end{bmatrix}$ as a linear combination of the vectors in S where a_1 and a_2 are scalars.

4. Determine whether the following set of vectors is linearly independent or dependent.

(i)
$$\begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 4 \\ 2 \\ -2 \end{bmatrix}$$

(ii)
$$\begin{bmatrix} 1 \\ 1 \\ 3 \\ 3 \end{bmatrix}, \quad \begin{bmatrix} -1 \\ 1 \\ -1 \\ -3 \end{bmatrix}, \quad \begin{bmatrix} 2 \\ 5 \\ 9 \\ 6 \end{bmatrix}$$

5. Consider the set of vectors
$$\begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix}$$
Is this set of vectors a basis for $\mathbb{R}^3$.

6. What is the rank of the matrix A where $A = \begin{bmatrix} 1 & 2 & 0 & 1 & 1 \\ 2 & 4 & 1 & 3 & 0 \\ 3 & 6 & 2 & 5 & 1 \\ -4 & -8 & 1 & -3 & 1 \end{bmatrix}$

7. Show that the transformation $T : \mathbb{R}^2 \to \mathbb{R}^3$ given as $T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 + 3x_2 \\ 3x_2 + 2 \\ 2x_1 - x_2 \end{bmatrix}$ is not linear.

8. Show that the transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ given as $T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 - |x_2| \\ x_1 + 2x_2 \end{bmatrix}$ is not linear.

9. If we define $T : \mathbb{R}^2 \to \mathbb{R}^2$ by $T(\mathbf{x}) = A\mathbf{x}$. Find the image of $\mathbf{v}$ under T if $A = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$.

10. Check whether the transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ given as $T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} 3x_1 + x_2 \\ -2x_1 - 4x_2 + 3x_3 \\ 5x_1 + 4x_2 - 2x_3 \end{bmatrix}$ is not linear or not.